

The background of the slide is a photograph of a pod of orcas (killer whales) swimming in clear blue water. The orcas are seen from below, with their dark bodies and white underbellies visible. They are swimming in various directions, creating a sense of movement and social interaction.

Behavioral development, learning, and cognition

IB 431 – Behavioral Ecology
Fall 2025
October 28th 2025

Outline

- I. Review next proposal assignment – full first draft of proposal
- II. Behavioral development, learning, and cognition
- III. Discuss Alcock 1972

Full first draft of proposal

- Due Friday November 7th at 11:59pm
- 3 pages, single spaced (okay if you got a bit over, but stay below 4 pages)
- Full description on Canvas

Full first draft of proposal

- (1) Title
- (2) Introduction
- (3) Hypotheses and predictions
- (4) Methods
- (5) Figure(s) of predicted results
- (6) Significance
- (7) References

Full first draft of proposal

- For the Hypotheses and predictions, Methods, and Figure(s) of predicted results, you can re-use the material from your first submissions of these sections
- Please try to incorporate any feedback the instructors left on your first submissions

Full first draft of proposal

(1) Title

(2) Introduction

(3) Hypotheses and predictions

(4) Methods

(5) Figure(s) of predicted results

(6) Significance

(7) References

Possible structure for introduction

- Introduce general concept
 - Last sentence – transition into general research gap
- Discuss general research gap
- Introduce your system
- Discuss research gap from above in context of your system
- Provide overview of study and state questions / hypotheses

Example: What is the effect of predation risk on activity in squirrels?

- Understanding the impacts of predation risk on phenotype is important and can help us better understand the selective forces shaping prey behavior...
- However, our understanding of the impacts of predation risk on prey activity remains limited...
- Squirrels are a strong model system for exploring this...
- Specifically, it is unclear if food availability has an impact on squirrel response to predation risk...
- Here, I provided squirrels with different diets and explored their response to auditory cues of predators...

Some ideas for significance section

- How will this study help us better understand this system/organism?
- How will this work further our understanding of evolution and ecology more generally?
- What future studies could this work lead to?
- Are there any broader applications; i.e., is it relevant to conservation, resource management, public health, industry, etc.?
- Is this topic of interest to the broader public - for instance, is this an enigmatic species? An economically important species? Could the results be used to develop community outreach activities?

Full first draft of proposal

- Title – feel free to be creative
- References – include at least 5 – but you may need more depending on how much background you set up, methods you need to cite, etc.
- Can use any citation style you like as long as you are consistent

Full first draft of proposal

- See Canvas site for more resources (all linked in the “Full first draft of proposal” assignment page)
- Full proposal description with more tips for writing introduction and significance
- Science writing lecture from earlier this semester
- Sample proposals
- Useful science writing phrases

Full first draft of proposal

- This is just the first draft – don't stress on making it perfect by the deadline
- You will be receiving feedback from your peers and then given a chance to revise before submitting your second draft
- Instructors will be lenient on grading for the first draft as long a good effort is made and all of the necessary sections are present
- The second draft is where we will be more critical

Outline

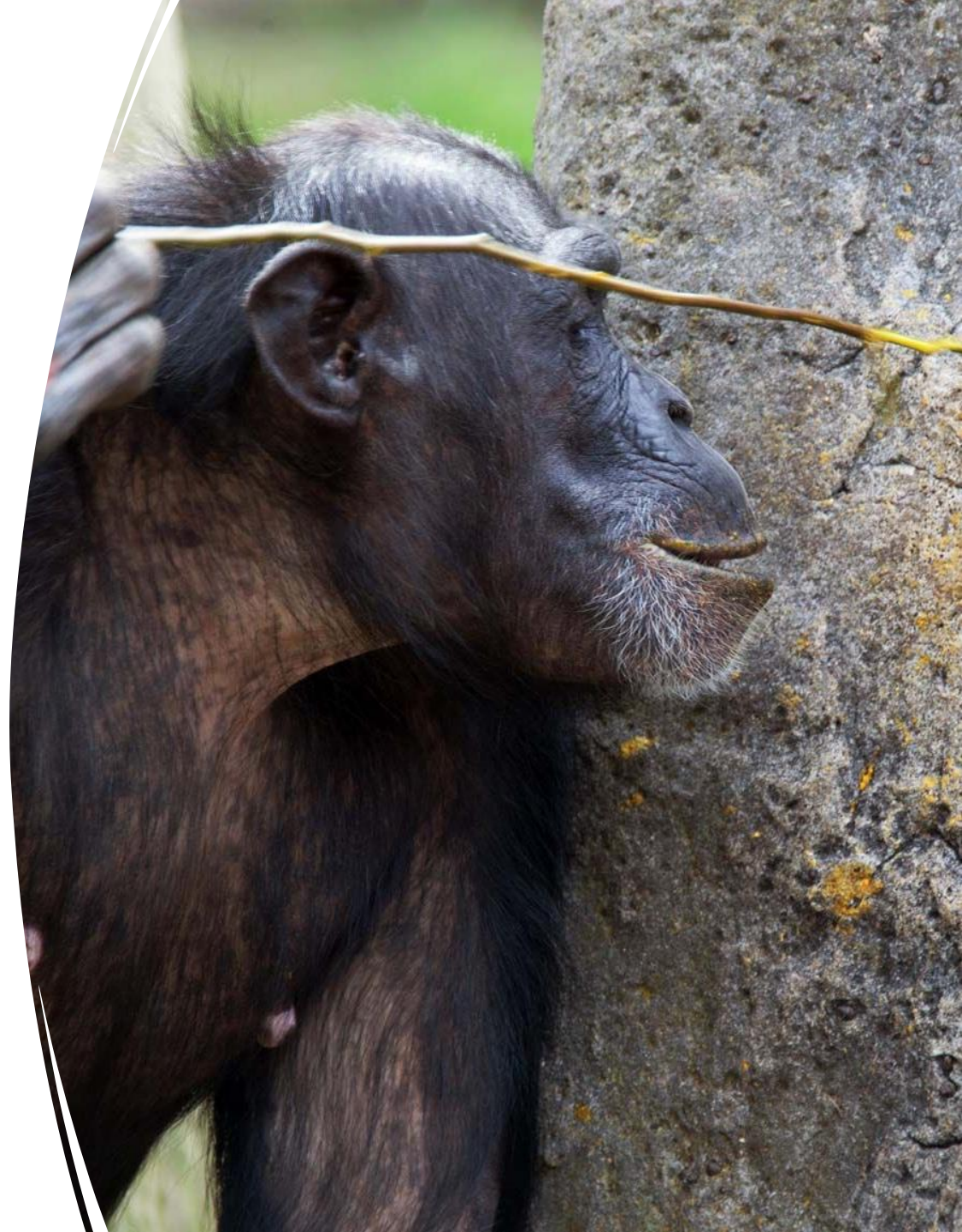
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Animal cognition and intelligence

- How can we define intelligence? What characteristics should be considered? Are there different forms of intelligence across different species?



- **Innate** = behavior is genetically determined
- **Learning** = relatively permanent change in behavior as a result of experience
 - Learning is a mechanism of phenotypic plasticity
- **Memory** = recalling what has been learned
- **Cognition** = mechanisms by which animals acquire, access, store and act on information from their environment
(Shettleworth 1998)



Innate behaviors

- Some behaviors are present without any prior experience or learning
- **Fixed action patterns** = behavioral sequence that is driven by instinct
 - Spider webs
 - Aggressive behavior in stickleback fish



How do animals learn?

- Single stimulus — e.g., red stick repeatedly dropped into an animal's cage, does the animal turn its head more or less with each subsequent drop?



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How do animals learn?

- Single stimulus — e.g., red stick repeatedly dropped into an animal's cage, does the animal turn its head more or less with each subsequent drop?
 - **Habituation:** animal becomes less sensitive to stimuli over time
 - **Sensitization:** animal becomes more sensitive to stimuli over time



Imprinting

- Individual forms strong social bond with first individual it encounters (typically a parent)
- Occurs during early critical period of development
- Can lead to issues if imprinting does not occur with parent



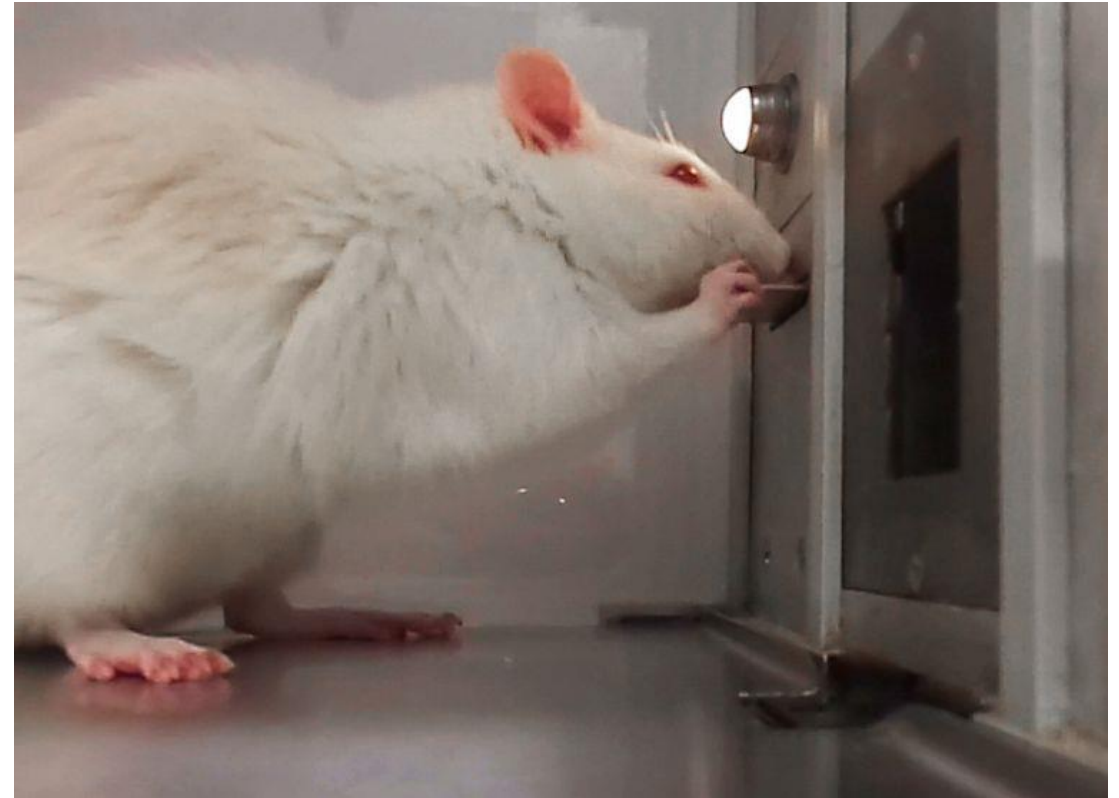
Associative learning

- Classical conditioning (Pavlovian)
 - Individual learns to associate two stimuli involuntarily
 - Biologically relevant and neutral stimuli
 - Neutral stimuli alone triggers response expected for the biological stimuli



Associative learning

- Operant conditioning
 - Voluntary response to stimuli reinforces behavior
 - Pressing lever provides food reward, rat will continue pressing lever



Rat Park experiment

- Rats will get learn to prefer morphine water over control water
- But – studies only done in empty cages; what if you gave rats other stimuli?



Rat Park experiment

- Given enriched environment (other rats, toys, large cage) – rats prefer normal water over drug water



Rat Park experiment

- Gene expression related to neurotransmitter pathways was altered
- Social interaction activated inhibitory neurons in the amygdala, reducing activation of pathways driving drug craving



Social learning

- **Local enhancement** = animal is drawn to where others are
- **Social facilitation** = presence of others increases behavior (e.g., use of red vs. blue doors in great tits; Aplin et al 2015)
- **Imitation** = observer acquires a novel response via observing a demonstrator



Teaching

- Orcas teach hunting strategies to future generations
- Stranding strategy = purposefully get washed ashore to hunt seals or sea lions resting on the shore
- Learned behavior that is only present in certain populations



Hoelzel 1990

Teaching

- Older meerkats teach juveniles how to hunt, particularly when foraging for more dangerous prey items like scorpions



Thornton 2008

Evolution of learning

- Under what conditions should learning evolve?
 - **Environment rarely changes** = best to avoid costs of learning and pass on information genetically
 - **Environment often changes** = learning not as useful because what was learned is irrelevant in the next situation
 - Intermediate levels of environmental predictability should lead to learning being favored

Evolution of learning

- Under what conditions should learning evolve?
- **Adaptive specialization hypothesis** = more stressful conditions may favor increased learning



Roth et al 2012

Evolution of learning

- Under what conditions should learning evolve?
- **Adaptive specialization hypothesis** = more stressful conditions may favor increased learning
- Chickadees in colder environments had better spatial memory, more developed hippocampus, greater caching propensity



Roth et al 2012

Cognitive bias

- Is this glass half-empty or half-full?



Animals exhibit cognitive bias

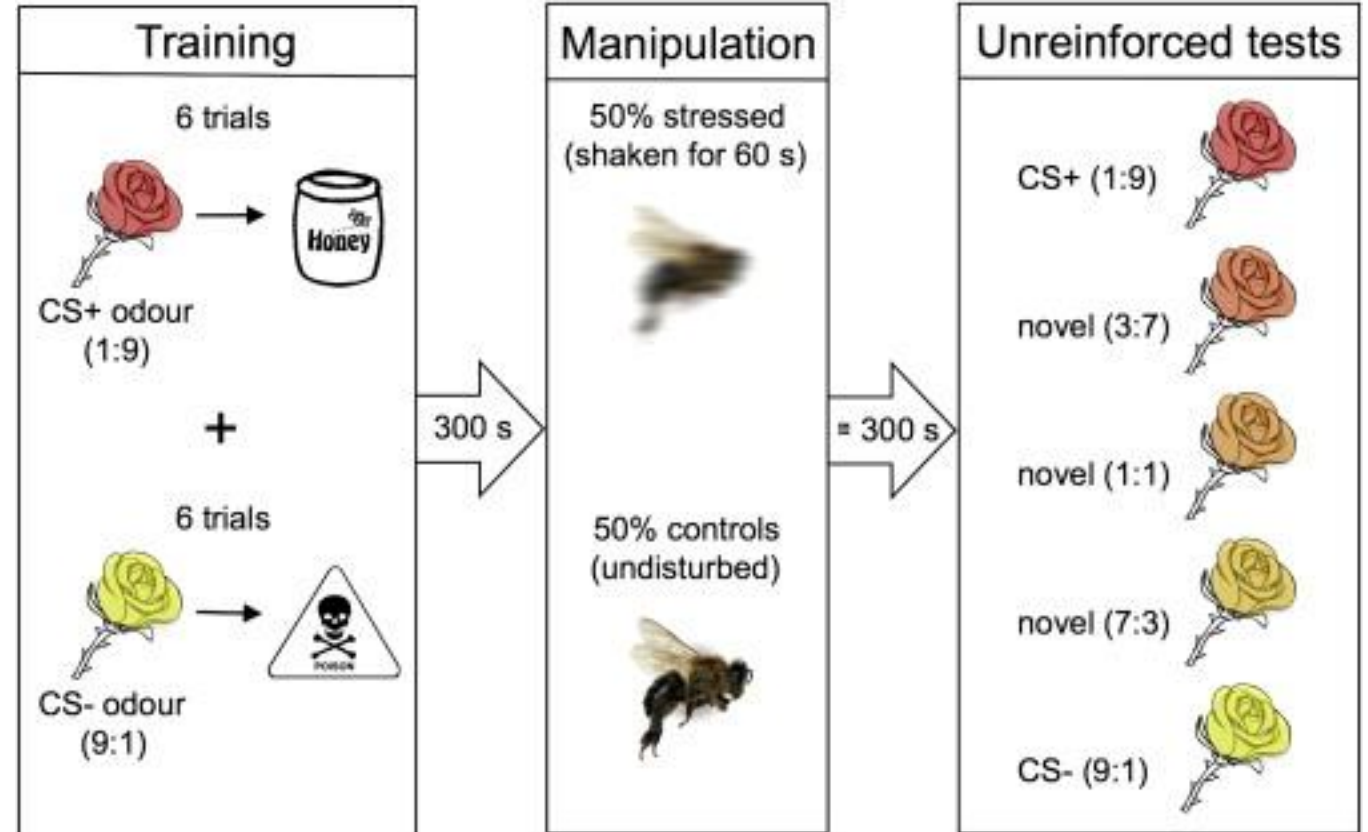
- Macaques were trained to touch small lines to get food reward; large lines led to mildly aversive stimulus
- Then given ambiguous lines
- Monkeys that had recently experienced positive enrichment touches ambiguous lines more than those that had health treatment



Bethell et al 2012

Even present in insects

- Honeybees that experienced simulated predator attack were more likely to classify ambiguous stimulus as a threat



Memory in wasps

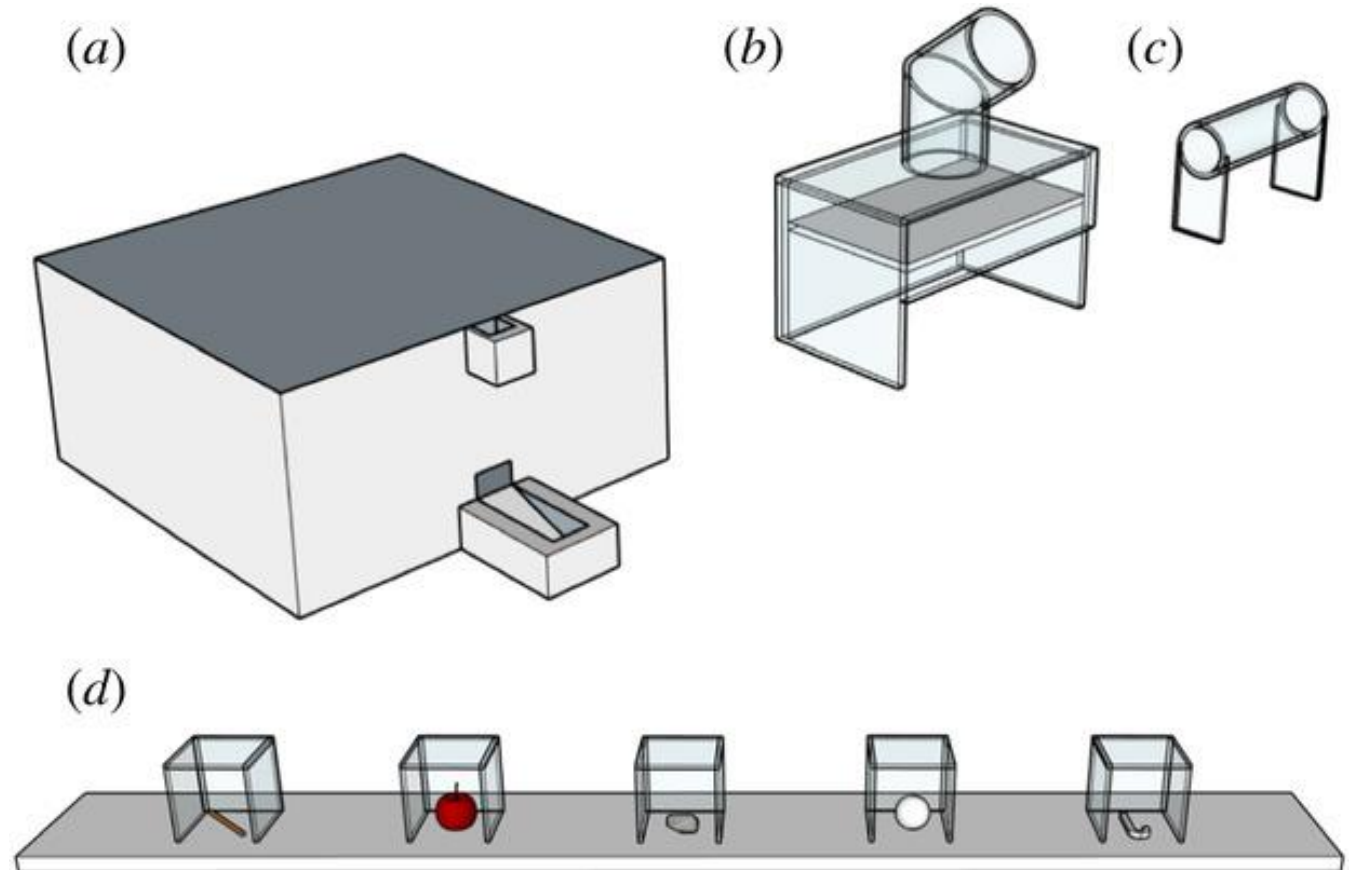
- Wasps remember faces and have complex dominance hierarchy based on observations of fights among pairs of wasps



Baracchi et al 2015

Planning ahead

- Crows were able to remember different solutions to puzzles and stored correct tools for future use



Memory and understanding of fairness

- Ravens trained to exchange low-quality for high-quality food item
- Some experimenters were “unfair”
- Ravens remember and don’t interact with them next time even a month later



Müller et al 2017



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